

REVIEW

Hypertension and the Brain-Heart Interconnection

Chronic Hypertension and the Brain's Waste Drainage System

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ABSTRACT: An important system for brain waste clearance comprises a series of interconnected fluid and tissue compartments: the cerebrospinal fluid compartment, the glymphatic system, and the dura with the meningeal lymphatic vasculature, which drains to the cervical lymphatics. Although exchanges between these compartments are increasingly well documented, the mechanisms underlying their crosstalk and especially the regulation of brain waste clearance efficiency—including the role of immune surveillance along this pathway—remain incompletely understood. This interconnected pathway, designated the glymphatic-meningeal-cervical-lymphatic axis, is critically dependent on the cerebral vasculature, both structurally and functionally. Yet given this intimate vascular dependence, the effects of chronic hypertension—one of the most prevalent chronic diseases worldwide—on the structural and functional integrity of this axis remain remarkably understudied. This review examines each component of this pathway in physiological sequence, evaluating the evidence for hypertension-related dysfunction at each node and identifying where critical gaps remain.



Key Words: blood vessels ■ cerebrospinal fluid ■ chronic disease ■ glymphatic system ■ hypertension

The brain clears waste through several pathways and biochemical processes. One such pathway consists of a series of functionally interconnected fluid and tissue compartments: the cerebrospinal fluid (CSF) compartment, the glymphatic system,¹ and the dura with the meningeal lymphatic vasculature (mLV), draining to the cervical lymphatic system^{2–5} (Figure). Together, these constitute a brain waste clearance route—increasingly referred to as the glymphatic-meningeal-cervical-lymphatic axis—whose disruption has been associated with neurodegenerative diseases including Alzheimer disease^{7–10} and cerebral small vessel disease (cSVD).^{11–14} The glymphatic model is one of several proposed frameworks for brain waste clearance. Alternatives include intramural periarterial drainage (IPAD) and models in which mixing accounts for brain solute clearance; these frameworks have been reviewed in detail elsewhere.^{15–18} We adopt the glymphatic model here because it provides the most developed account of the full brain-to-lymphatic pathway. Glymphatic transport is also strongly state-dependent, varying with sleep,¹⁹ anesthesia,^{20,21} circadian phase²² and body posture.²³ Sleep disruption²⁴—including

obstructive sleep apnea²⁵—is itself closely linked to hypertension. These factors act as system-wide modulators rather than discrete anatomic nodes; detailed discussion is beyond the scope of this review and has been provided elsewhere.²⁶

Glymphatic brain waste clearance is thought to be mediated through bulk-flow driven CSF influx along peri-arterial spaces into the interstitial fluid (ISF) compartment.¹ CSF-ISF exchange is facilitated by AQP4 (aquaporin-4) water channels expressed on astrocytic endfeet, although reduced transport and waste clearance can still occur in the absence of AQP4.¹ Waste-laden fluid exits the ISF along perivenous channels—including along bridging veins—which traverse the arachnoid membrane at discrete arachnoid cuff exit (ACE) points.²⁷ At these sites, the perivenous spaces transferring waste effluent are directionally distinct from periarterial influx, allowing waste transport into the dural interstitial space. There, excess fluid and waste solutes are taken up by the mLV and drained to the cervical lymph nodes or gain direct access to the systemic circulation^{3,28} (Figure).

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Nonstandard Abbreviations and Acronyms

ACE	arachnoid cuff exit
AQP4	aquaporin-4
Aβ	amyloid beta
BBB	blood-brain barrier
CSF	cerebrospinal fluid
cSVD	cerebral small vessel disease
DCE-MRI	dynamic contrast-enhanced magnetic resonance imaging
IPAD	intramural periarterial drainage
ISF	interstitial fluid
mLV	meningeal lymphatic vasculature
MMP	matrix metalloproteinase
PBM	parenchymal border macrophage
PVS	perivascular space
SHR	spontaneously hypertensive rat
SHRSP	spontaneously hypertensive stroke-prone rat

Given this intimate vascular dependence, pulsatility^{29,30} and vasomotion^{31–33} are key determinants of glymphatic function, rendering the system inherently vulnerable to vascular disease. Yet the consequences of chronic arterial hypertension—among the most prevalent vascular disorders worldwide³⁴—for the structural and functional integrity of the glymphatic-meningeal-cervical-lymphatic axis remain remarkably understudied. This review examines each component of the pathway in physiological sequence, assessing at each node the structural and functional consequences of chronic hypertension for brain waste clearance.

HISTORICAL CONTEXT

For much of scientific history, the brain parenchyma has been known to be devoid of lymphatic vasculature. This led to the long-standing perception that cerebral metabolic waste clearance relied on processes such as passive diffusion into the CSF acting as a sink³⁵ and transport across the blood-brain barrier (BBB).^{36,37} In the 19th century, Rudolf Virchow and Charles Philippe Robin independently described perivascular spaces (PVS) surrounding cerebral blood vessels.^{38,39} The anatomist William His Sr. suggested that the PVS might carry lymph-like fluid, though this remained speculative.⁴⁰ Still, it has long been known that CSF played a role as an active third circulation, moving directionally as a central nervous system fluid system distinct from blood. Cushing described the third circulation as a CSF circulation system that was not passive but actively driven and with physiological significance for brain homeostasis and intracranial pressure regulation.⁴¹

Several decades later, Rennels et al²⁹ subsequently showed that the brain's PVS act as conduits for CSF, by demonstrating rapid perivascular uptake of horse-radish peroxidase injected into the CSF compartment of cats and dogs. In parallel, Cserr and Ostrach laid the conceptual and experimental foundation for CSF-ISF mediated brain clearance. They measured solute clearance rates by administering radiolabeled tracers into the brain parenchyma or CSF and reported that solute transport reflected bulk flow of ISF toward the CSF sink and cervical lymphatics.^{42–45}

Despite these insights, the mechanistic framework for a brain clearance pathway on the brain parenchymal side of the BBB remained incomplete until Iliff et al¹ proposed the glymphatic system in 2012. The term glymphatic reflected both its dependence on glial cells and its functional analogy to the peripheral lymphatic system. Using 2-photon microscopy in live mice, they demonstrated a coordinated sequence of (1) convective peri-arterial CSF influx, (2) CSF- ISF exchange facilitated by AQP4 water channels, and (3) AQP4-dependent perivenous solute waste outflow.¹ Notably, AQP4, which is highly enriched on perivascular astrocytic endfeet,⁴⁶ proved functionally impactful: AQP4-knockout mice showed significantly impaired Aβ (amyloid beta) clearance compared with wild-type controls,^{1,19} a finding later supported by pharmacological AQP4 blocking studies.⁴⁷ The glymphatic system transport was later extended to the whole brain in vivo using dynamic contrast-enhanced magnetic resonance imaging (DCE-MRI) with intrathecal Gd-tagged tracers, confirming peri-arterial influx and bulk-flow driven glymphatic transport at macroscopic scale.⁴⁸

THE GLYMPHATIC-MENINGEAL-CERVICAL LYMPHATIC AXIS

The glymphatic system is best understood as a functional model: a brain-wide perivascular network that serves as a pathway for CSF to exchange with ISF to clear waste from the brain (Figure). The perivascular channel network associated with all cerebral vessels, is thought to serve as the structural scaffold of the glymphatic system and the key drivers of transport are predominantly vascular. Arterial pulsatility is required for CSF movement and entry into the perivascular spaces.^{1,30} The dependence of glymphatic transport on pulsatility supports the notion that the CSF-mediated glymphatic transport is advectively (bulk-flow) driven. Another driver of glymphatic transport and waste clearance is vasomotion, which is slow, rhythmic oscillations in the arterial tone.^{49,50}

The original glymphatic model of Iliff et al¹ described perivenous channels as the primary outflow route—likely reflecting the pressure gradient that drives fluid from peri-arterial inflow toward peri-venous drainage—but the

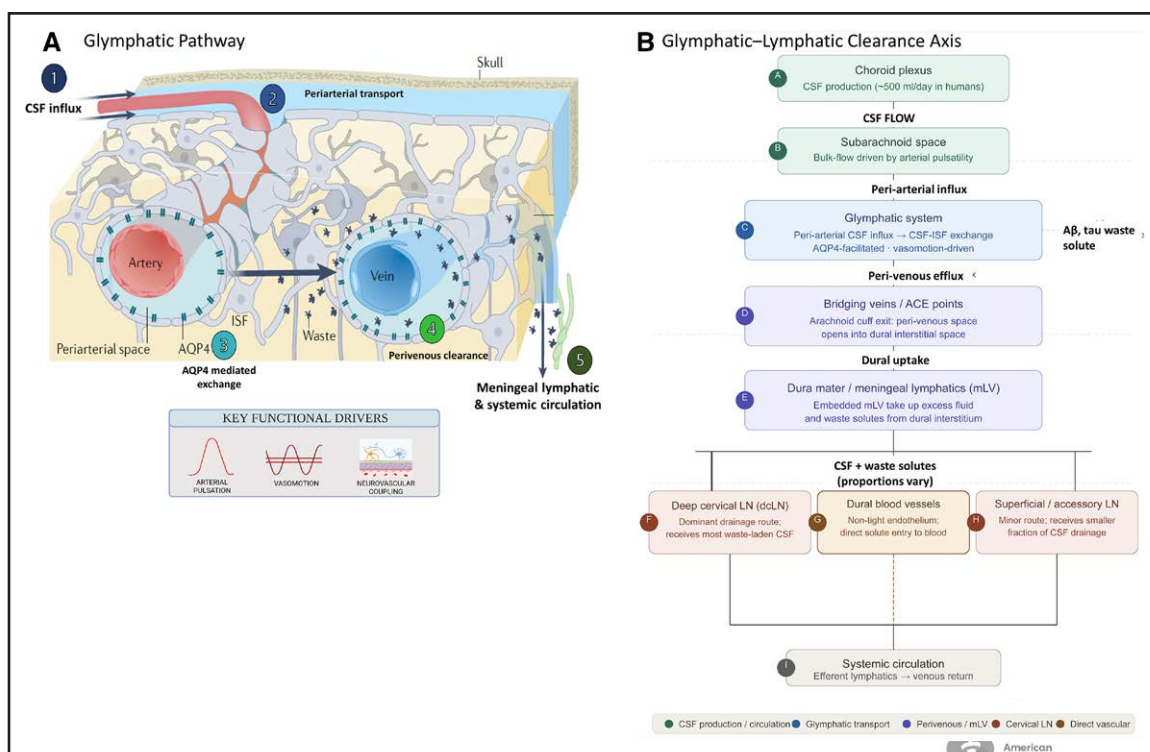


Figure. Overview of the glymphatic–lymphatic clearance system.

A, Schematic representation of the glymphatic–lymphatic pathway, illustrating cerebrospinal fluid (CSF) influx along periaxonal spaces, AQP4 (aquaporin-4)-mediated exchange with interstitial fluid, perivascular clearance, and drainage through meningeal lymphatic vessels to the systemic circulation. Key physiological drivers of glymphatic transport, including arterial pulsatility, vasomotion, and neurovascular coupling, are highlighted and contribute to cerebrospinal fluid movement and solute clearance. **B**, Conceptual overview of the glymphatic–lymphatic clearance axis, depicting the sequential movement of CSF and solutes from the choroid plexus and subarachnoid space through the glymphatic system, bridging veins/arachnoid cuff exit (ACE) points, meningeal lymphatics, deep cervical lymph nodes, and systemic circulation. Adapted from Lohela TJ et al⁶ with permission. Copyright ©2022, Springer Nature.

precise mechanism(s) by which waste solutes exit perivascular spaces remained poorly defined. A pivotal advance came in 2015, when the mLV was (re)discovered and systematically mapped in the mouse using genetic reporter tools.^{2,3} These vessels were found to run along the dural sinuses and the meningeal arteries^{2,3} and cranial nerve sheaths⁵¹ as they exit the skull. Experiments with intraparenchymal tracer injections established that mLV remove waste solutes from the brain, draining them to the cervical lymph nodes. Over the last decade the connections between the glymphatic system, the mLV and cervical lymphatic system have been mapped and are collectively designated the glymphatic–meningeal–cervical lymphatic axis.¹⁶

The sections that follow trace this pathway in physiological sequence with a focus on the impact of chronic hypertension. We begin with CSF production and circulation, establishing CSF as the substrate of the entire system. We then examine peri-arterial influx and CSF-ISF exchange, including the modulatory roles of AQP4. Finally, we follow the waste-laden fluid through peri-venous outflow channels, towards the meningeal and cervical lymphatic drainage routes to the systemic circulation. At each node, we examine how chronic hypertension might

disrupt the process and what the consequences are for brain waste clearance.

CSF PRODUCTION, COMPOSITION AND CIRCULATION

The major portion of CSF is produced by the choroid plexuses where tight junctions of the epithelial cells form the blood-CSF barrier.^{52,53} Choroidal CSF secretion is an active process driven by apically polarized ion transporters—principally $\text{Na}^+/\text{K}^+/\text{ATPase}$, NKCC1 , and HCO_3^- transporters—that couple ion translocation to fluid secretion.^{52,53} NKCC1 is particularly noteworthy: it transports ions and water simultaneously but acts as a reversible transporter whose net direction is determined by the combined electrochemical gradients of Na^+ , K^+ , and Cl^- .^{52,53} Under normal conditions, the reversed apical polarity of $\text{Na}^+/\text{K}^+/\text{ATPase}$ and the relatively high intracellular Cl^- in the epithelial cells support net outward secretion, but this balance can shift under pathological conditions.

Choroidal blood flow is among the highest per gram of any tissue in the body. Unlike cerebral capillaries,

choroidal capillaries are fenestrated,⁵⁴ exposing the basolateral epithelial surface directly to a plasma ultrafiltrate that is then actively processed by the transepithelial ion transport machinery. Choroidal blood flow appears to lack the robust pressure-autoregulation characteristic of cerebral parenchymal vessels, making CSF production more directly dependent on systemic perfusion pressure^{53,55}—though the precise autoregulatory capacity of choroidal vessels remains incompletely characterized. Notably, the choroid plexus receives dense sympathetic innervation from the superior cervical ganglia, targeting both the choroidal vascular smooth muscle cells and the secretory epithelium.^{56,57} Stimulation of the sympathetic fibers reduces net CSF production by up to 30%, while sympathectomy increases it by a comparable magnitude indicating that the choroid plexus operates under substantial tonic sympathetic inhibitory tone under normal conditions.⁵⁷ Any sustained reduction in choroidal perfusion—as may occur with vasoconstrictors or potentially antihypertensive medication—might therefore alter CSF output.

COMPOSITION OF CSF

CSF is an actively secreted fluid whose ionic composition broadly resembles plasma but differs substantially as protein and glucose content is significantly lower.⁵⁸ CSF also contains solutes from the brain parenchyma including A β and tau, now established as diagnostic biomarkers for Alzheimer's disease.^{59–62} This reflects CSF's role as a sink: by maintaining low metabolic waste concentrations relative to ISF, it sustains the concentration gradient driving diffusion and convective waste export from the parenchyma. The extent to which waste-laden perivenous efflux mixes with the arterial CSF influx compartment—and thereby limits net clearance efficiency through recirculation—remains incompletely understood and represents an important unresolved question.

CSF CIRCULATION

CSF secreted by the choroid plexus enters the subarachnoid space; from there a fraction— $\sim 20\%$ ⁶³—is drawn into periarterial influx channels and undergoes glymphatic CSF-ISF exchange, while the remainder traverses the subarachnoid compartment without entering the parenchyma. The subarachnoid space thus supplies both the periarterial influx route and, separately, CSF that reaches the exit routes without traversing the parenchyma. Within the brain, periarterial influx and perivenous efflux remain directionally distinct, the perivenous spaces carrying ISF-derived effluent rather than subarachnoid CSF. Both the glymphatic-processed and the bypassing fractions can exit via ACE points connecting to the dural interstitial space and meningeal lymphatics²⁷ as well as other routes^{45,64,65} (Figure [B]). The importance of normal CSF production for glymphatic function is supported by the

observation that CSF leakage, such as from dural tears or iatrogenic maneuvers, drastically reduces glymphatic solute influx and waste clearance.⁶⁶ A recently recognized cellular regulator of CSF influx is the parenchymal border macrophage population residing at the CSF-CNS interface.^{67,68}

CHRONIC HYPERTENSION AND CSF HOMEOSTASIS

Production

Although no published postmortem study has formally examined choroidal vascular structure specifically in hypertensive subjects, the existing aging literature is substantially confounded by hypertension given that more than 50% of community-dwelling individuals over 72 have hypertension. Therefore, choroid plexus pathology documented in elderly postmortem series—including calcification, hyalinization, and lipid deposition—likely reflects, at least in part, the cumulative effects of chronic elevated blood pressure rather than aging per se.⁶⁹ Formally disentangling age-related from hypertension-specific vascular remodeling of the choroid plexus would require dedicated postmortem studies with careful stratification by lifetime blood pressure history—which to our knowledge have not been performed. In preclinical studies, most evidence is derived from rat models of chronic hypertension, particularly the spontaneously hypertensive rat (SHR), which develops early onset ventriculomegaly independently of blood pressure, since antihypertensive treatment does not reverse ventricular enlargement once established.⁷⁰ Within the SHR model, however, the relationship between hypertension and CSF production remains unresolved: 1 study reported normal intracranial pressure, CSF production, and normal resistance to outflow in SHR compared with normotensive controls,⁷¹ while another reported significant CSF hypersecretion emerging specifically after the onset of hypertension at 7 weeks of age.⁷² This discrepancy remains unexplained.

Beyond production rate, the antihypertensive pharmacological landscape adds to this knowledge gap. The drug classes most prescribed for hypertension—angiotensin receptor blockers, angiotensin-converting enzyme inhibitors, and calcium channel blockers—have essentially no published literature examining their direct effects on choroidal CSF production, composition, or blood-CSF barrier integrity, despite strong mechanistic rationale for such effects via the renin-angiotensin-mineralocorticoid axis at the choroid plexus.^{73–75}

Circulation

CSF transport in the subarachnoid space and along the pial arteries is bulk-flow driven and occurs in synchrony with arterial pulsation.⁷⁶ Using particle tracking

velocimetry, acute pharmacologically induced hypertension was shown to alter the shape of the arterial wall pulsation by introducing an additional fast-contraction phase that generates significant backflow and thereby reduces net forward CSF PVS flow, despite the artery diameter itself remaining unchanged.⁷⁶ Equivalent studies have not been performed in models of chronic hypertension, and it remains unknown whether chronic vascular remodeling produces a distinct flow signature. However, CSF flow dynamics have been characterized using DCE-MRI and computational fluid dynamics analysis in adult spontaneously hypertensive stroke-prone rats (SHRSP).¹² This work demonstrated that solute speed along pial arteries such as the middle cerebral artery was reduced, in adult SHRSP rats compared with normotensive WKY controls.¹²

In summary, the evidence for choroid plexus involvement in chronic hypertension is incomplete and contradictory. Human postmortem evidence specifically addressing choroidal vascular pathology in chronic hypertension is limited to aging. Finally, the pharmacological dimension of this gap is striking: the antihypertensive drug classes most widely prescribed have essentially no published data on their effects on choroidal function or barrier integrity.

PERI-ARTERIAL INFLUX

The Perivascular Space

The perivascular space is a compartment that accompanies cerebral blood vessels from the pial surface arteries through penetrating arterioles and into the deeper parenchyma of the brain. Its boundaries are formed internally by the vessel wall and externally by the glial limitans. The space is populated by several resident cell populations—parenchymal border macrophages (PBMs), pericytes, and fibroblasts—each contributing to the regulation of perivascular fluid dynamics, extracellular matrix composition, and immune surveillance.⁷⁷ At the capillary level the space becomes virtual, with the basement membrane serving as the primary conduit for fluid and solute transport.⁷⁸ Together this perivascular channel network constitutes the structural scaffold of the glymphatic system, through which CSF enters the brain parenchyma and exchanges with ISF.

PERI-ARTERIOLEAR DRIVERS

Glymphatic influx begins at the arterial perivascular space and is bulk-flow driven.¹ Animal studies have demonstrated the importance of this directly: unilateral ligation of the internal carotid artery reduces arterial pulsatility and glymphatic influx, while pharmacological agents that increase arterial pulsatility correspondingly enhance influx.³⁰ However, this analysis

primarily captured cardiac-frequency pulsations and did not assess slower oscillations such as vasomotion. In addition, potential effects of carotid ligation on downstream lymphatic drainage were not evaluated. Studies in both rodents and humans confirm that tracer influx from the CSF follows a centripetal pattern, beginning at the brain surface and later extending to deeper regions.^{1,48,79,80} It is important to note that due to technical limitations, the concept that peri-arterial inflow is bulk-flow driven throughout the brain's perivascular system remains inferred rather than directly demonstrated. Nevertheless, advective CSF transport is well established in the subarachnoid space and along large pial arteries in both rodents and humans.^{76,81} The advective force arises primarily from vascular pulsatility, with additional contributions from other physiological drivers including respiration.^{30,76,82} Note that we use the term driver here to denote the mechanisms generating the pressure gradients that move fluid through the perivascular compartments, rather than the source of the fluid (CSF) itself. We also distinguish 2 senses of pulsatility used in this literature. Local arterial wall motion refers to the cyclical displacement of the vessel wall, measured directly in preclinical studies; perivascular CSF movement is synchronized to the cardiac cycle, and CSF velocity matches the speed of the moving wall, which has been interpreted as identifying wall motion as the proximate driver of perivascular flow.⁷⁶ Transmitted pressure pulsatility refers instead to the amplitude of the intraluminal pressure wave—e.g., from the aorta reaching the cerebral circulation, and is the variable captured clinically. Where we describe pulsatility as a driver of perivascular flow we mean wall motion; where we describe augmented pulsatility in chronic hypertension we mean transmitted pressure.

Another vascular driver of glymphatic influx is vasomotion—slow, rhythmic oscillations in arterial tone arising primarily from intrinsic smooth muscle automaticity at ≈ 0.1 Hz.^{83,84} Using in vivo 2-photon microscopy in awake mice, Van Veluw et al (2020) demonstrated that vasomotion amplitude correlates with the rate of clearance of interstitial solutes along pial arteries, and that experimentally increasing vasomotion amplitude via evoked neurovascular responses correspondingly increased clearance.³¹ Notably, evoked neurovascular reactivity was impaired in mice with cerebral amyloid angiopathy, with a corresponding reduction in clearance rate—directly linking vasomotor dysfunction to impaired solute drainage in a disease-relevant model.³¹ More recent work has extended this concept to infraslow (≈ 0.02 Hz) vasomotor oscillations driven by norepinephrine release from the locus coeruleus during NREM sleep, which also propel CSF along perivascular spaces.⁸⁵ Further support comes from experiments showing that optogenetically induced arterial wall motion, in the absence of neuronal activation, is sufficient to enhance glymphatic influx.⁸⁵

A caveat applies to all of the physiological drivers: cardiac pulsation and vasomotion act on the same vessel wall at different frequencies, and no experimental preparation has succeeded in isolating one from the other. Interventions that alter cardiac pulsatility inevitably alter vascular tone as well. The strongest experimental support for pulsatility as a driver of net perivascular flow therefore comes from direct velocimetry, in which particle displacement is measured in phase with the arterial wall over the cardiac cycle,⁷⁶ and the strongest support for vasomotion comes from optogenetically imposed wall motion generating influx in the absence of neuronal activation. Both demonstrate that arterial wall motion can produce net transport.³³

How vasomotion produces net directional transport remains unresolved. Computational simulations have suggested that arterial wall oscillations alone are insufficient to generate net axial flow, with modelled rigid body motions producing local fluid swirling that could instead enhance mixing.^{31,86,87} Moreover, long-wavelength vasomotion waves have been shown to propagate bidirectionally along cortical arteries in awake mice.⁸⁸ We note that, bidirectional wall motion does not preclude net solute transport, which can arise from dispersion and from network geometry rather than from the direction of individual waves,⁸⁷ and more recent models demonstrate that vasomotion can generate net directional flow in asymmetrical branching networks.⁸⁹ The relative contributions of mixing versus genuine advection therefore likely depend on local vascular geometry and remain to be resolved experimentally.

PBMs are also important for glymphatic influx. Specifically, PBM depletion impairs CSF influx into the glymphatic system, with 2 potentially distinct mechanisms proposed: a direct role for arterial-associated (LYVE1⁺) PBMs in regulating arterial wall motion, and a slower process involving reduced MMP (matrix metalloproteinase) activity and consequent extracellular matrix accumulation in the perivascular space.⁶⁸ The relative timescales and contributions of these 2 mechanisms have not been fully resolved. Importantly, perivascular macrophages exist as a cell population in the human brain, although their precise location within the PVS has not been definitively established.⁹⁰ Transcriptomic studies indicate that this population is altered in human Alzheimer disease.⁶⁸

PERI-ARTERIAL INFLUX AND CHRONIC HYPERTENSION

Systemically, hypertension induces structural and functional remodeling across the vascular tree, affecting both large arteries and microcirculation. In small resistance vessels, chronic elevations in blood pressure lead to inward remodeling characterized by narrowing of the vascular lumen and an increased media-to-lumen ratio, resulting in impaired vasodilatory capacity and increased

vascular resistance.^{91,92} Larger arteries instead undergo outward hypertrophic remodeling, with medial thickening expanding the outer vessel dimension.⁹² In regards to the brain we focus here on cSVD, for which chronic hypertension is a major—though not the only—risk factor, and in which the pattern of remodeling differs again: cSVD is associated with a maladaptive phenotype of arterial widening and elongation (dolichoectasia), which relates to lacunar stroke and cSVD burden, and has been interpreted as reflecting segmental arteriolar disorganization.⁹³ Notably, this form of remodeling is accompanied by enlarged perivascular spaces, one of the defining imaging features of cSVD, indicating that the perivascular compartment expands along with the remodeling vessel rather than being compressed by it.^{94,95}

Arterial (eg, the aorta) stiffening in chronic hypertension augments pulsatility which is transmitted to the cerebral arteries.^{96,97} At first glance this appears difficult to reconcile with reduced glymphatic influx, since arterial pulsatility is a principal driver of perivascular CSF movement.^{1,30} As defined above, these are not equivalent measures, and in chronic hypertension they may diverge: reduced arterial compliance both permits a larger pressure pulse to reach the cerebral microcirculation and would be expected to diminish the wall motion available to drive flow, although this has not been directly measured in cerebral arteries in vivo. In acute hypertension, a distortion of the wall motion waveform was sufficient to generate retrograde flow and reduce net forward perivascular transport even though mean arterial diameter was unchanged.⁷⁶ Whether an analogous waveform distortion develops in chronic hypertension is unknown. Moreover, cerebral vasoreactivity—a clinical marker of cerebrovascular responsiveness—is impaired in chronic hypertension.^{98,99} Notably, isolated cerebral arteries from SHRSP rats exhibit a markedly higher propensity for spontaneous vasomotion *ex vivo* compared with normotensive WKY arteries¹⁰⁰—a finding whose functional implications for glymphatic transport remain unresolved, as discussed previously.

In humans, the Framingham Heart Study cohort (n=3710) found that older age, hypertension, higher systolic and diastolic blood pressure, and smoking were all associated with a higher burden of enlarged PVS.⁹⁵ These findings should be interpreted with caution, since the pathophysiological significance of enlarged PVS remains unresolved, as does their causal relationship to chronic hypertension. Although epidemiological studies consistently associate hypertension with greater enlarged PVS burden, PVS enlargement does not resolve with blood pressure lowering, and direct case-control comparison has not reproduced the association.¹⁰¹ Consistent with this, enlarged PVS are only rarely detected on MRI in SHRSP rats¹⁰²—which may reflect the limited spatial resolution achievable in rodents, but may equally indicate that chronic hypertension is not

sufficient to produce them. Enlarged PVS may therefore be a marker of small vessel disease that is associated with, rather than caused by elevated blood pressure. A further consideration is that an enlarged or apparently unchanged PVS may coexist with functionally reduced transport if vessel wall remodeling or accumulation of extracellular material narrows the effective cross-section available for flow.

The functional status of PBMs—the cell population critical for maintaining perivascular patency and CSF flow—remains unexplored in chronic hypertension. PBM depletion attenuates neurovascular and cognitive dysfunction in angiotensin II and deoxycorticosterone acetate salt models,^{103,104} but these studies did not assess CSF influx or clearance, nor distinguish among PBM subtypes, so whether hypertension affects the LYVE1⁺ arterial-associated PBM population implicated in perivascular fluid dynamics is unknown (Table).

CSF-ISF EXCHANGE: THE INTERSTITIAL COMPARTMENT

When tracers are administered into the CSF compartment, they appear in the brain’s ISF, demonstrating that solutes—carried alongside CSF—are transported across the perivascular-interstitial interface. This process is referred to as CSF-ISF exchange (Figure). The passage of solutes from the perivascular space into the ISF is molecular size-dependent.¹ The physical forces

responsible for this transport are both bulk flow (advection) and diffusion, with an additional contribution from dispersion.¹⁵ To clarify these terms: advection refers to transport driven by a pressure gradient, in which solutes are carried by the bulk movement of fluid.^{12,15} Diffusion is solute movement driven by an electrochemical concentration gradient, independent of fluid movement. Dispersion—specifically Taylor dispersion—arises from velocity gradients within the flowing fluid: solutes in faster-moving fluid layers spread relative to those in slower layers, producing effective mixing that exceeds what diffusion alone would generate.^{15,16,87} The relative contributions of advection and diffusion to CSF-ISF exchange remain debated.^{42,55,105} Diffusion alone cannot account for the observed rates of solute transport across the distances involved.^{42,55} In disease states, additional restrictions on diffusion—arising from extracellular matrix remodeling, cellular swelling, or altered ISF composition—may further impair transport, though the specific contribution of impaired diffusion relative to impaired advection in conditions such as chronic hypertension has not been directly resolved.

The transport of solutes and fluid from the perivascular space into the ISF is facilitated by AQP4 water channels expressed on astrocytic endfeet. Genetic deletion of AQP4 reduces glymphatic CSF tracer influx by ≈25% to 65% depending on the study and metric used.^{1,22} Acute pharmacological inhibition with the selective AQP4 blocker AER-271 produces a more modest reduction of ≈20%.⁴⁷ The functional effectiveness of AQP4 depends

Table. Knowledge Gaps Along the Glymphatic–Meningeal–Cervical Lymphatic Axis in Chronic Hypertension

Node level	Knowledge gap	Priority	Evidence
CSF production (choroid plexus)	No postmortem human data on hypertension-related vascular remodeling of choroidal vasculature.	High	Aging data
CSF production (choroid plexus)	Contradictory findings on CSF production rate in SHR rats. Discrepancy unresolved.	High	Preclinical
CSF production (choroid plexus)	No published data on effects of widely prescribed antihypertensive drugs on choroidal CSF production.	High	No data
CSF circulation	Effect of long-term arterial remodeling on pulsatile flow waveform and net CSF transport unknown.	High	No data
Perivascular space (peri-arterial influx)	Extracellular matrix composition of the hypertensive perivascular space is essentially unstudied.	Medium	No direct data
Perivascular space (peri-arterial influx)	PBMs in chronic hypertension is unexplored.	Medium	No data
CSF-ISF exchange (glymphatic transport)	Functional implications of increased ex vivo vasomotion in SHRSP cerebral arteries are unresolved.	Medium	Only ex vivo data
CSF-ISF exchange (glymphatic transport)	Relative contributions of impaired advection vs impaired diffusion to CSF-ISF exchange in chronic hypertension have not been resolved.	Medium	Preclinical data
Perivenous outflow (clearance)	No data on whether chronic hypertension alters bridging vein structure or ACE point integrity.	Medium	Preclinical data
mLV (clearance)	No dedicated studies characterizing structural or functional changes in dural mLV in chronic hypertension.	High	No data
Cervical lymph nodes (clearance)	No study has tracked brain waste clearance through the full glymphatic-mLV-cervical lymph node axis under chronic hypertension.	Critical	No data
	Cervical lymph nodes may undergo hypertension-related fibrosis and structural remodeling, independently elevating outflow resistance.	High	No data

ACE indicates arachnoid cuff exit; CSF, cerebrospinal fluid; ECM, extracellular matrix; ISF, interstitial fluid; mLV, meningeal lymphatic vasculature; PBM, parenchymal border macrophage; SHR, spontaneously hypertensive rat; and SHRSP, spontaneously hypertensive stroke-prone rat.

critically on its polarization to the perivascular endfeet. Displacement toward the glial cell soma such as observed in aging¹⁰⁶ substantially reduces its contribution to perivascular water transport. AQP4 depolarization has been documented in aging and neurodegenerative states including in human brain tissue in Alzheimer disease.¹⁰⁷ MRI studies in AQP4 knockout mice have demonstrated reduced water exchange across the blood-CSF interface,^{108,109} consistent with AQP4's primary role in facilitating transcellular water flux.

Vasomotion and neurovascular coupling are critically relevant as vascular drivers of CSF-ISF exchange,^{31,33,85} through distinct mechanisms—vasomotion through rhythmic arterial wall oscillations that drive perivascular fluid movement, and neurovascular coupling through activity-dependent vasodilation that modulates local flow dynamics. Both have been shown to influence the rate of CSF-ISF exchange and waste clearance.^{16,31,33}

CHRONIC HYPERTENSION AND CSF-ISF EXCHANGE

Direct evidence for impaired CSF-ISF exchange in chronic hypertension is limited. DCE-MRI studies in SHR and SHRSP rats demonstrated reduced global solute influx compared with normotensive WKY controls,^{12,13} supporting impaired CSF-ISF exchange at the perivascular-vascular interface. Several converging mechanisms likely contribute. Neurovascular coupling is impaired in chronic hypertension through endothelial dysfunction and oxidative stress.¹¹⁰ Critically, NVC impairment has been demonstrated in hypertensive patients without symptomatic cerebrovascular disease,⁹⁸ suggesting it is an early consequence of sustained elevated blood pressure rather than a late manifestation of established vascular injury. In addition, AQP4 polarization to the perivascular astrocytic endfeet is disrupted in SHRSP rats,¹² which may further impair the facilitated water transport that supports CSF-ISF exchange. The state of the extracellular matrix in the hypertensive perivascular space—and its potential contribution to impaired solute exchange—remains essentially unstudied and represents an important knowledge gap. Taken together, these observations suggest a mechanism linking impaired CSF-ISF exchange to the perivascular enlargement described above. If AQP4 depolarization and an altered extracellular matrix reduce the transfer of CSF from the perivascular space into the ISF, fluid entering the perivascular compartment may stagnate rather than move onward. On this account—proposed previously as a glymphopathy model of cSVD¹¹¹—enlarged PVS would represent a downstream consequence of impaired perivascular transport. This hypothesis remains untested and would require serial imaging linking glymphatic function to PVS enlargement within the same animals or patients. It is also important to highlight that the effect of

chronic hypertension on brain waste clearance has not been documented.

PERI-VEIN OUTFLOW AND WASTE CLEARANCE FROM THE BRAIN

Among the most prominent functions of the glymphatic-mLV-cervical lymphatic pathway is the clearance of brain waste products, including A β and tau. In the original paper by Iliff et al,¹ A β was observed to deposit along large cerebral veins, giving rise to the concept that brain waste clearance occurs along perivenous channels. Perivenous solute waste clearance has been significantly validated and conceptually extended by an elegant study by Smyth et al,²⁷ who demonstrated that CSF and solutes exit the brain along perivascular channels associated with bridging veins. Critically, this efflux occurs at discrete anatomic loci termed ACE points.²⁷ The ACE points are discontinuities in the arachnoid barrier layer where bridging veins penetrate through to the dura, creating direct conduits between the subarachnoid space and the dural interstitial space.²⁷ Notably, at ACE points, the bridging vein endothelium converts from a BBB phenotype to a peripheral vascular phenotype with looser endothelial connections that permits molecular exchange between the blood and the perivenous space. In humans, this is evidenced by gadolinium-based contrast leaking from bridging veins into the subarachnoid space and parasagittal dura following intravenous administration.²⁷ ACE points have been characterized in detail along the dorsal surface of the rodent brain and in humans, and analogous exit sites are thought to exist along the ventral surface.¹¹² In humans, arachnoid granulations represent an additional exit route, abutting the dura and creating conduits between the ISF, subarachnoid CSF, and the dural interstitial space.¹¹³

Although perivenous efflux is supported by convergent evidence across species, brain waste clearance remains an area of active investigation and the efflux routes are not fully mapped. Solutes leaving the ISF are transported along cerebral and deeper veins, converging on dorsal bridging veins and ACE points or on ventral exit routes. Most experimental evidence derives from CSF-side tracer administration, in which a substantial proportion of tracer bypasses the parenchyma entirely, so that what is observed at the exit may not represent brain ISF-derived waste. Perivenous efflux is moreover one of several exit routes, and no study has apportioned the relative contribution of each within the same preparation. Whether these routes are redundant, and whether their relative use shifts in disease states including chronic hypertension, is unknown. It also remains unresolved whether all perivenous efflux is ultimately handled by the meningeal lymphatics. Solutes reaching the dural interstitium may be taken up by mLV, absorbed directly into dural blood

vessels, or pass into skull bone marrow channels,¹¹⁴ and the relative capacity of these routes has not been quantified.

Within the dura, the embedded mLV take up excess fluid and waste solutes draining from the brain. This has been demonstrated and independently reproduced across multiple laboratories. The mLV are present on both the dorsal and ventral dura, and their drainage pattern to the cervical lymph nodes has been mapped.^{2–5,112} There is broad consensus that waste solutes originating from CSF-ISF exchange—referred to as dirty CSF—drain preferentially to the deep cervical lymph nodes, whereas CSF that bypasses the glymphatic system spill over to other cervical lymph node groups, including the superficial and accessory cervical lymph nodes.⁴

The full brain-to-mLV-to-cervical lymph node drainage pattern in humans remains incompletely understood. Novel therapeutic strategies aimed at enhancing brain-to-lymph drainage via lymphovenous bypass^{115,116} are currently under investigation and may yield important new insights into the drainage patterns in humans.

THE EFFECT OF CHRONIC HYPERTENSION ON BRAIN WASTE CLEARANCE

To the best of the authors' knowledge, no studies have directly documented the effect of chronic hypertension on waste clearance via the brain-meningeal lymphatic-cervical lymphatic axis. This represents a critical gap. Based on existing evidence from SHRSP and SHR rat models demonstrating impaired CSF flow dynamics and reduced glymphatic uptake, it is reasonable to infer that downstream outflow to the cervical lymph nodes will be similarly compromised. However, interpreting such studies is inherently complex, since chronic hypertension is a systemic disease affecting all organ systems—including the brain, the meningeal vasculature, and the peripheral lymphatic system. Attributing impaired brain-to-cervical lymph node drainage specifically to upstream glymphatic failure versus downstream resistance within the cervical lymph nodes themselves would therefore require careful experimental design. The cervical lymph nodes are themselves susceptible to hypertension-related pathological remodeling, including fibrosis and structural changes that independently increase outflow resistance¹¹⁷—potentially confounding the interpretation of tracer-based drainage studies.

Chronic hypertension is also associated with BBB disruption, though the evidence is less clean than often assumed and intersects with brain clearance in ways that complicate interpretation. In humans, a 7T DCE-MRI study found no difference in BBB leakage between hypertensive and normotensive participants,

although leakage correlated with systolic, diastolic, and mean arterial pressure as continuous measures.¹¹⁸ Pre-clinical evidence is heterogeneous: BBB permeability changes are frequently reported in SHRSP rats maintained on salt-loading protocols that accelerate stroke and carry blood pressure-independent effects on endothelial function^{119,120}—making it difficult to attribute the observed leakage to chronic elevated pressure alone. Importantly, a compromised BBB provides a route for solute egress directly into the vasculature, independent of perivascular transport. Tracer clearance may therefore appear preserved or even accelerated in the presence of BBB breakdown, despite impaired glymphatic function—an effect we have observed in a model of chronic lymphatic drainage impairment.²⁸ Disentangling these upstream and downstream contributions will be essential for understanding the full impact of chronic hypertension on the glymphatic-meningeal-cervical lymphatic axis and represents an important priority for future investigation.

CHRONIC HYPERTENSION, CEREBRAL SMALL VESSEL DISEASE, AND THE GLYMPHATIC AXIS

Chronic arterial hypertension affects more than one billion people worldwide³⁴ and remains the leading modifiable risk factor for stroke, cognitive impairment, and dementia. It is a common risk factor for ischemic and hemorrhagic stroke, dementia including Alzheimer's disease, as well as the condition linking stroke and dementia—cSVD—a syndrome characterized on neuroimaging by white matter hyperintensities, lacunar infarcts, microbleeds, and enlarged perivascular spaces.¹²¹ These lesions accumulate over years, increase with age, and other vascular risk factors including diabetes, hyperlipidemia, smoking, and other factors among which hypertension is an important and modifiable vascular risk factors. Longitudinal data confirm that progression continues even in patients treated with anti-hypertensive medication to lower versus usual targets, suggesting that blood pressure lowering alone is insufficient to halt the underlying microvascular injury.¹²² This concept is reinforced by the LACI-2 trial,¹²³ which evaluated 2 repurposed drugs that target microvascular dysfunction rather than blood pressure reduction per se, and reduced SVD complications without affecting blood pressure, reflecting a growing recognition that the therapeutic target in cSVD must extend well beyond simply treating high blood pressure.

The mechanisms linking sustained hypertension to white matter injury and cognitive decline remain incompletely understood, but the evidence reviewed here suggests that impairment of the glymphatic-meningeal-cervical-lymphatic axis may be an important and underappreciated contributor. Acutely elevated blood pressure

alters arterial wall motion waveform, increasing retrograde perivascular CSF flow and reducing net glymphatic influx in mice⁷⁶; whether chronic hypertension with structural arterial remodeling produces equivalent or more severe disruption of perivascular flow dynamics remains to be established. Impaired neurovascular coupling, AQP4 depolarization, and potential perivascular extracellular matrix remodeling might converge in chronic hypertension to reduce CSF-ISF exchange efficiency and reduce the clearance of waste solutes including A β and tau from the brain. Enlarged perivascular spaces—a key imaging feature of cSVD especially early in disease—may reflect this functional failure: not simply anatomic widening, but a structural signature of chronically impaired perivascular fluid transport.^{124,125} In this framework, cumulative hypertensive burden may not only drive vascular wall injury but likely also progressive waste clearance failure, with neuroinflammation and toxic protein accumulation as downstream consequences. Prospective studies directly linking glymphatic function metrics—measurable in humans by noninvasive MRI techniques—to cSVD lesion burden and cognitive trajectories in hypertensive cohorts would test this hypothesis and could establish glymphatic dysfunction as a therapeutic target in its own right.

KNOWLEDGE GAPS AND FUTURE DIRECTIONS

This review reveals that while the glymphatic–meningeal–cervical lymphatic axis is increasingly well characterized under physiological conditions, evidence for how chronic hypertension disrupts each node of this pathway remains fragmentary. Several interconnected gaps stand out (Table).

At the level of CSF production, 2 fundamental problems remain unsolved. Human postmortem data on choroidal vascular pathology in hypertension are limited to aging studies. Equally striking is the pharmacological blind spot: the antihypertensive drug classes most widely prescribed—ARBs, ACE inhibitors, and calcium channel blockers—have essentially no published data on their effects on choroidal function, despite compelling mechanistic rationale via the renin-angiotensin-mineralocorticoid axis at the choroid plexus.

Moving downstream, the studies which have revealed how acute hypertension disrupts arterial wall pulsation waveform and reduces net perivascular CSF flow have not been performed in chronic hypertension models where structural remodeling has developed over time. Moreover, the functional status of parenchymal border macrophages—a cell population critical for maintaining perivascular patency and CSF flow—is similarly unexplored in chronic hypertension (Table).

Most critically, no study has tracked brain waste clearance along the full glymphatic–meningeal

lymphatic–cervical lymph node axis under conditions of chronic hypertension. DCE-MRI evidence from SHRSP rats establishes upstream glymphatic failure, but whether this translates to impaired cervical lymph node drainage has not been demonstrated (Table).

Finally, virtually all functional evidence reviewed here derives from preclinical rodent models. Translational studies in humans—leveraging noninvasive MRI platforms validated for measuring glymphatic and meningeal lymphatic function *in vivo*—are urgently needed to determine whether the glymphatic impairment documented in hypertensive rats is recapitulated in hypertensive patients, and whether it tracks with disease duration, treatment status, and the burden of cerebral small vessel disease. Such studies would provide the foundation for rationally designed therapeutic strategies targeting the glymphatic–lymphatic axis as a modifiable driver of hypertension-related cognitive impairment.

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